Foundations Notebook

Your Name

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Chapter 1

Mathematical Foundations

1.1 Linear Algebra

1.1.1 Vector Spaces

Definition 1.1.1. A vector space V over a field $\mathbb{F}$ is ...

1.1.2 Why This Matters

Linear algebra underpins signal spaces, optimization, and estimation theory.

Chapter 2

Electrical Engineering Foundations

2.1 Electro Magnetism

2.2 Radar

Radar: Radio Detection And Ranging. Is simply the transmission and receiving of Radio waves. From these radio waves it is possible to determine many things about the incident target.

Simple Model

A transmitter is to be thought as a isotropic radiator with the propogation pattern described by equation 2.1.

$$P_d = \frac{P_t}{4\pi r^2} \tag{2.1}$$

Where P_d is the power desnity at some radius, P_t is the power transmitted and r is the radius.

$$R = \frac{c\delta t}{2} \tag{2.2}$$

2.2.1 SAR

2.3 Signals and Systems

2.3.1 Sampling

2.3.2 LTI Systems

An LTI system satisfies:

- Linearity
- Time invariance